

SEN-107 Fundamental Data Structures
Assessment Activity SEN-107:00020
Understanding and Using Recursion
Last update: 9 January 2025

Instructions

General Instructions

Write a program that solves the problem described below. Your program should read from the standard input and write to the standard output. We have provided example input and output files. When running your program on the example input file, it should produce the same output as in the example output file. While this example is a good starting point, you should test your program thoroughly, with multiple different inputs too. Your program should correctly handle all possible cases specified below. Correctness will be part of your assessment score. Please also follow the coding standards in *C_CodingStandards.pdf*. Violating the coding standards will lower your assessment score.

Submit your **C source code** only. Your source code must be a single plain text file. Do not submit code as a doc file, pdf file, IDE project, or anything else. We must be able to compile and run your code on my computer, without any warnings or errors.

For convenience, we have provided a template for you, which already contains the skeleton code that handles the input-output. You may implement your solution using any data structure you learn from lectures (or beyond). However, you are **not** allowed to use other external libraries, such as C++ Standard Template Library (STL).

Assignment-Specific Instructions

For this assessment, you should submit a single file called *matrilinealDescendant.c*; as stated above, we have already provided a template for this file. The example input and output files are *matrilinealDescendant.in* and *matrilinealDescendant.out*, respectively. Note that you can adapt your code from Lab 2 (*matrilineal.c*) for this assessment.

Problem Statement: Finding and Printing Descendants

The setting is the same as Lab 2 except that, instead of querying for mother, daughters and sisters, we are interested in the following two queries of the descendants:

- Print longest chain of descendants
 - If there are multiple chains of descendants that have the same (maximum) length, you must output all of them.
 - The chains may be printed in any order; however, the order within each chain must strictly follow the descendant tree (i.e., a mother must always come before her daughter).
- Print all the descendants of some woman in the tree

Input Description

The first line of the input contains a number T denoting the number of operations.
Each of the next T lines denotes one of the two events.

- The first line will be a string, which is the name of the woman at the root of the tree. (This is considered an add operation.)

- If the line starts with “Add”, it indicates an add operation. It will be followed by two strings, the name of the women and the name of her mother, respectively.
- If the line starts with “Descendant”, it is asking to find a women’s descendants. This is followed by the name of the women to query.
- If the line is “Max”, it is asking for a chain of descendants with maximum length.

You can assume that each input name has length at most 20.

Output Description

After each operation, the program should print the following in a new line:

- Add operation:
 - If the person is (i) added as a child of someone who has 10 children already or (ii) the mother is not found, then print “Unsuccessful”.
 - Otherwise, if the person is successfully added, print “Added”.
- For “Descendant” operation, print the corresponding names. When there are multiple such names, they should be separated by whitespaces and they can be printed in any order. In the case that no such person exists (e.g., the queried woman is not in the tree or the root of the tree has no mother), print “Not Found”.
- For “Max” operation, print all longest chains. Each chain should appear on its own line, with names printed from the top down and separated by whitespace.

Example Input & Output

Input	Output
19	Added
Emma	Added
Add Jane Emma	Added
Add Jenny Jane	Added
Add Linda Jane	Added
Add Tina Jenny	Added
Add Nancy Tina	Added
Add Kate Tina	Nancy Kate Tina
Descendant Jenny	Not Found
Descendant Kate	Added
Add Susan Emma	Added
Add Martha Emma	Added
Add Ann Martha	Added
Add Patty Martha	Added
Add Eliza Patty	Added
Add Fran Patty	Added
Add Vanessa Eliza	Emma Jane Jenny Tina Nancy
Max	Emma Jane Jenny Tina Kate
Add Dora Vanessa	Emma Martha Patty Eliza Vanessa
Max	Added
	Emma Martha Patty Eliza Vanessa Dora

Explanation: The final constructed tree is shown on the next page. This tree has a single longest chain of descendants: *Emma* → *Martha* → *Patty* → *Eliza* → *Vanessa* → *Dora* which is the answer to the final query. Before Dora was added, there were multiple longest chains, as shown in the tree below. The program above outputs each chain from left to right. Your program may output the chains in any order, as long as each chain follows the correct descendant order.

